

[BUG] Error for user-defined device `dtype` with the TF interface

<https://github.com/PennyLaneAI/pennylane/issues/2105>

ROW 69

[BUG] Error for user-defined device dtype with the TF interface #2105

New issue

Closed

#2120



antalszava opened on Jan 18, 2022

Expected behavior

No error arising related to dtype mismatch when setting custom dtype's on the device and using TF.

Actual behavior

The following error is raised with a QNode that takes floats as input and a TF loss:

```
InvalidArgumentError: cannot compute AddN as input #10(zero-based) was expected to be a float tensor but
```

Additional information

There might be a conversion happening in the backward pass (potentially in the custom gradient implementation). When omitting the use of `SparseCategoricalCrossentropy`, the gradient can be obtained without an issue.

Source code

```
import tensorflow as tf
import numpy as np
import pennylane as qml

nwires = 5
dev = qml.device("lightning.qubit", wires=nwires)
dev.C_DTYPE = np.complex64
dev.R_DTYPE = np.float32

@qml.qnode(dev, interface='tf', diff_method='adjoint')
def circuit(weights, features=np.zeros(nwires)):
    for i in range(nwires):
        qml.RX(features[i], wires=i)
        qml.RX(weights[i], wires=i)
    return [qml.expval(qml.PauliZ(0)), qml.expval(qml.PauliZ(1))]

np.random.seed(42)

ndata = 100
data = [np.random.randn(nwires).astype('float32') for _ in range(ndata)]
label = [np.random.choice([1, 0]).astype('int') for _ in range(ndata)]

loss = tf.losses.SparseCategoricalCrossentropy()

params = tf.Variable(np.random.randn(nwires).astype('float32'), trainable=True)
with tf.GradientTape() as tape:
    probs = [circuit(params, d) for d in data]
    loss_value = loss(label, probs)

grads = tape.gradient(loss_value, [params])
```

Tracebacks

No response

System information

Assignees

No one assigned

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Use `qml.math.cast` after `qml.math.convert_like` in `qml.gradients` to copy `dtype`
PennyLaneAI/pennylane

Notifications

Customize

Subscribe

You're not receiving notifications from this thread.

Participants



Name: PennyLane
Version: 0.21.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: <https://github.com/XanaduAI/pennylane>
Author:
Author-email:
License: Apache License 2.0
Location: /home/antal/xanadu/pennylane
Requires: numpy, scipy, networkx, retworkx, autograd, toml, appdirs, semantic_version, autoray, cachetools, psutil
Required-by: PennyLane-Lightning, PennyLane-Cirq, PennyLane-Orquestra, pennylane-qulacs, amazon-braket-pennylane-plugin
Platform info: Linux-5.11.0-44-generic-x86_64-with-glibc2.10
Python version: 3.8.5
Numpy version: 1.21.4
Scipy version: 1.7.3
Installed devices:
- lightning.qubit (PennyLane-Lightning-0.20.1)
- cirq.mixedsimulator (PennyLane-Cirq-0.19.0)
- cirq.pasqal (PennyLane-Cirq-0.19.0)
- cirq.qsim (PennyLane-Cirq-0.19.0)
- cirq.qsimh (PennyLane-Cirq-0.19.0)
- cirq.simulator (PennyLane-Cirq-0.19.0)
- orquestra.forest (PennyLane-Orquestra-0.15.0)
- orquestra.ibmq (PennyLane-Orquestra-0.15.0)
- orquestra.qiskit (PennyLane-Orquestra-0.15.0)
- orquestra.qulacs (PennyLane-Orquestra-0.15.0)
- qulacs.simulator (pennylane-qulacs-0.17.0.dev0)
- braket.aws.qubit (amazon-braket-pennylane-plugin-1.4.1.dev0)
- braket.local.qubit (amazon-braket-pennylane-plugin-1.4.1.dev0)
- honeywell.hqs (PennyLane-Honeywell-0.16.0.dev0)
- qiskit.aer (PennyLane-qiskit-0.18.0.dev0)
- qiskit.basicaer (PennyLane-qiskit-0.18.0.dev0)
- qiskit.ibmq (PennyLane-qiskit-0.18.0.dev0)
- aqt.noisy_sim (PennyLane-AQT-0.18.0)
- aqt.sim (PennyLane-AQT-0.18.0)
- projectq.classical (PennyLane-PQ-0.18.0.dev0)
- projectq.ibm (PennyLane-PQ-0.18.0.dev0)
- projectq.simulator (PennyLane-PQ-0.18.0.dev0)
- forest.numpy_wavefunction (PennyLane-Forest-0.18.0.dev0)
- forest.qvm (PennyLane-Forest-0.18.0.dev0)
- forest.wavefunction (PennyLane-Forest-0.18.0.dev0)
- microsoft.QuantumSimulator (PennyLane-qsharp-0.19.0)
- ionq.qpu (PennyLane-IonQ-0.17.0.dev0)
- ionq.simulator (PennyLane-IonQ-0.17.0.dev0)
- strawberryfields.fock (PennyLane-SF-0.20.0.dev0)
- strawberryfields.gaussian (PennyLane-SF-0.20.0.dev0)
- strawberryfields.gbs (PennyLane-SF-0.20.0.dev0)
- strawberryfields.remote (PennyLane-SF-0.20.0.dev0)
- strawberryfields.tf (PennyLane-SF-0.20.0.dev0)
- default.gaussian (PennyLane-0.21.0.dev0)
- default.mixed (PennyLane-0.21.0.dev0)
- default.qubit (PennyLane-0.21.0.dev0)
- default.qubit.autograd (PennyLane-0.21.0.dev0)
- default.qubit.jax (PennyLane-0.21.0.dev0)
- default.qubit.tf (PennyLane-0.21.0.dev0)
- default.qubit.torch (PennyLane-0.21.0.dev0)

Existing GitHub issues

☒ I have searched existing GitHub issues to make sure the issue does not already exist.



antalszava added **bug** on Jan 18, 2022



antalszava changed the title **[BUG] Error for customly set device `dtype` with the TF interface** **[BUG] Error for user-defined device `dtype` with the TF interface** on Jan 18, 2022



yitchen-tim on Jan 18, 2022

Adding info about the bug:

It only happens with big number of `ndata`. For example, there is no bug when `ndata=10` instead of 100. The value of `ndata` which starts triggering the bug also depends on the random seed. When the seed is 42, the bug happens when `ndata` is bigger than 14 on my machine.

 **antalszava** added **bug** on Jan 18, 2022

 **antalszava** changed the title **[BUG] Error for customly set device `dtype` with the TF interface** **[BUG] Error for user-defined device `dtype` with the TF interface** on Jan 18, 2022



yitchen-tim on Jan 18, 2022

Adding info about the bug:

It only happens with big number of `ndata` . For example, there is no bug when `ndata=10` instead of 100. The value of `ndata` which starts triggering the bug also depends on the random seed. When the seed is 42, the bug happens when `ndata` is bigger than 14 on my machine.



josh146 on Jan 19, 2022

Member ...

Thanks [@yitchen-tim](#) for the details! Although I find it very odd that it is dependent on the size of the features 🤔

[@antalszava](#) is it like the case that there are more `float64` assumptions hardcoded in somewhere?



 **antalszava** mentioned this on Jan 22, 2022

[Use `qml.math.cast` after `qml.math.convert\_like` in `qml.gradients` to copy `dtype`](#) #2120

 **antalszava** linked a pull request that will close this issue [Use `'qml.math.cast'` after `'qml.math.convert\_like'` in `'qml.gradients'` to copy `'dtype'`](#) #2120 on Jan 24, 2022

 **antalszava** closed this as **completed** in #2120 on Jan 25, 2022

Use `qml.math.cast` after `qml.math.convert_like` in `qml.gradients` to copy dtype #2120

<> Code

Merged antalszava merged 12 commits into `master` from `compute_vjp_use_cast_like` on Jan 25, 2022

Conversation 6 Commits 12 Checks 0 Files changed 4 +114 -5



antalszava commented on Jan 22, 2022 · edited

Contributor

Context:

`qml.math.cast_like` attempts to copy the dtype of a tensor whereas `qml.math.convert_like` only uses the type of the tensors during conversion

The `qml.gradients` module uses `qml.math.convert_like`. There may, however, be cases where the dtype would need to be copied, otherwise unexpected behaviour arises.

Description of the Change:

Benefits:

Possible Drawbacks:

Related GitHub Issues:

[#2105](#)



use `cast_like` instead of `convert_like` in the gradient pipelines to pr... 8d84766



github-actions bot commented on Jan 22, 2022

Contributor

Hello. You may have forgotten to update the changelog!

Please edit [doc/releases/changelog-dev.md](#) with:

- A one-to-two sentence description of the change. You may include a small working example for new features.
- A link back to this PR.
- Your name (or GitHub username) in the contributors section.



Reviewers

josh146

Jaybsoni

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[BUG] Error for user-defined device dtype ...

Notifications

Customize

Subscribe

You're not receiving notifications from this thread.

3 participants

